

NYC Shooting Incidents

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2025-04-12

- Data set is NYPD Shooting Incident Data(Historic) last updated on 4/26/2024 on catalog.data.gov.
- Data entries range from 1/1/2006 to 12/29/2023
- What boroughs have more incidents than others and how many total incidents are there across all boroughs?

```
#This code starts the cleaning process
nyshooting <- "https://data.cityofnewyork.us/api/views/833y-fsy8/rows.csv?accessType=DOWNLOAD"
nyshooting <- read.csv(nyshooting)

#Changing weird values to UNKNOWN so most data is not lost
sore_val <- c("", "(null)", "1028", "1020", "940", "224", "1022")

#Changing names of columns to easier names
nyshooting <- rename(nyshooting, ID_Key = "INCIDENT_KEY", Date = "OCCUR_DATE", Time = "OCCUR_TIME", Bor
                        Perp_Age = "PERP_AGE_GROUP", Perp_Sex = "PERP_SEX", Perp_Race = "PERP_RACE",
                        Vic_Age = "VIC_AGE_GROUP", Vic_Sex = "VIC_SEX", Vic_Race = "VIC_RACE", Precinct =
                        "PRECINCT", Jur_Code = "JURISDICTION_CODE") %>%
select(-"LOC_OF_OCCUR_DESC", -"LOC_CLASSFCTN_DESC", -"X_COORD_CD", -"Y_COORD_CD",
      -"Latitude", -"Longitude", -"Lon_Lat", -"ID_Key", -"Inci_Loc") %>%
mutate(across(where(is.character),
              ~ifelse(.x %in% sore_val, "UNKNOWN", .x)))

nyshooting <- na.omit(nyshooting)
nyshooting$Date <- as.Date(nyshooting$Date, format = "%m/%d/%Y")
nyshooting$Death <- as.logical(nyshooting$Death)
nyshooting$Jur_Code <- as.factor(nyshooting$Jur_Code)
nyshooting$Boro <- as.factor(nyshooting$Boro)
nyshooting$Precinct <- as.factor(nyshooting$Precinct)
nyshooting$Perp_Age <- as.factor(nyshooting$Perp_Age)
nyshooting$Perp_Sex <- as.factor(nyshooting$Perp_Sex)
nyshooting$Perp_Race <- as.factor(nyshooting$Perp_Race)
nyshooting$Vic_Age <- as.factor(nyshooting$Vic_Age)
nyshooting$Vic_Sex <- as.factor(nyshooting$Vic_Sex)
nyshooting$Vic_Race <- as.factor(nyshooting$Vic_Race)
#Variable Changes
#Date: character -> date
#Death: character -> logical
#All other variables: character -> factor

#Only 2 entries out of 28,562 entries contained NA values. Values are omitted
#Date allows for quantitative analysis over a selected time
#Logical for death illustrates a binary choice between survival and death of victim
```

#Factor variables allow for easier categorical variable analysis

```
head(nyshooting)
```

```
##      Date      Time      Boro Precinct Jur_Code Death Perp_Age Perp_Sex
## 1 2021-08-09 01:06:00  BRONX      40        0 FALSE  UNKNOWN  UNKNOWN
## 2 2018-04-07 19:48:00 BROOKLYN    79        0  TRUE   25-44        M
## 3 2022-12-02 22:57:00  BRONX      47        0 FALSE  UNKNOWN  UNKNOWN
## 4 2006-11-19 01:50:00 BROOKLYN    66        0  TRUE  UNKNOWN        U
## 5 2010-05-09 01:58:00  BRONX      46        0  TRUE   25-44        M
## 6 2012-07-22 21:35:00  BRONX      42        2 FALSE   18-24        M
##      Perp_Race Vic_Age Vic_Sex Vic_Race
## 1      UNKNOWN  18-24        M    BLACK
## 2    WHITE    HISPANIC  25-44        M    BLACK
## 3      UNKNOWN  25-44        M    BLACK
## 4      UNKNOWN  18-24        M    BLACK
## 5        BLACK    <18        F    BLACK
## 6        BLACK  18-24        M    BLACK
```

#Transformed data set nydeath of how many incidents occurred on each day

```
nydeath <- nyshooting %>%
```

```
  arrange(Date)
```

```
nydeath <- nydeath %>%
```

```
  group_by(Date, Death) %>%
```

```
  summarise(count = n(), .groups = "drop")
```

```
head(nydeath)
```

```
## # A tibble: 6 x 3
##   Date      Death count
##   <date>    <lgl> <int>
## 1 2006-01-01 FALSE     4
## 2 2006-01-01  TRUE     4
## 3 2006-01-02 FALSE     3
## 4 2006-01-02  TRUE     1
## 5 2006-01-03 FALSE     3
## 6 2006-01-03  TRUE     1
```

#Condensed bar chart where survival incidents stack on death incidents

#Total incidents for a given day are illustrated by the top point of survival bars

```
nydeath %>%
```

```
  ggplot(aes(x = Date, y = count, fill = Death)) +
```

```
  geom_col(position = "stack") +
```

```
  labs(title = "Daily Shooting Incidents in New York City",
```

```
        x = "Date", y = "Number of Incidents") +
```

```
  scale_x_date(date_labels = "%b %Y", date_breaks = "1 year") +
```

```
  scale_y_continuous(limits = c(0, 50)) +
```

```
  scale_fill_manual(values = c("TRUE" = "red", "FALSE" = "darkgreen"),
```

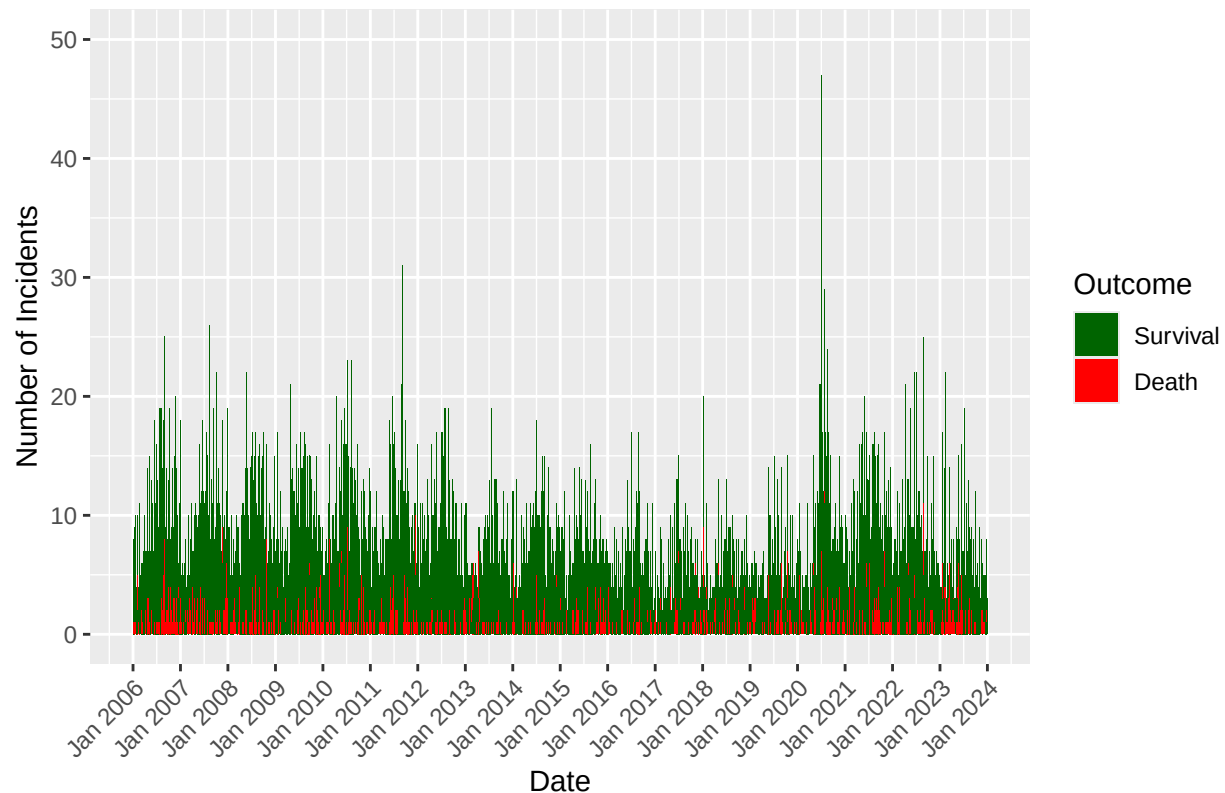
```
                      name = "Outcome",
```

```
                      labels = c("FALSE" = "Survival", "TRUE" = "Death")) +
```

```
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
```

```
  theme(plot.title = element_text(hjust = 0.5))
```

Daily Shooting Incidents in New York City

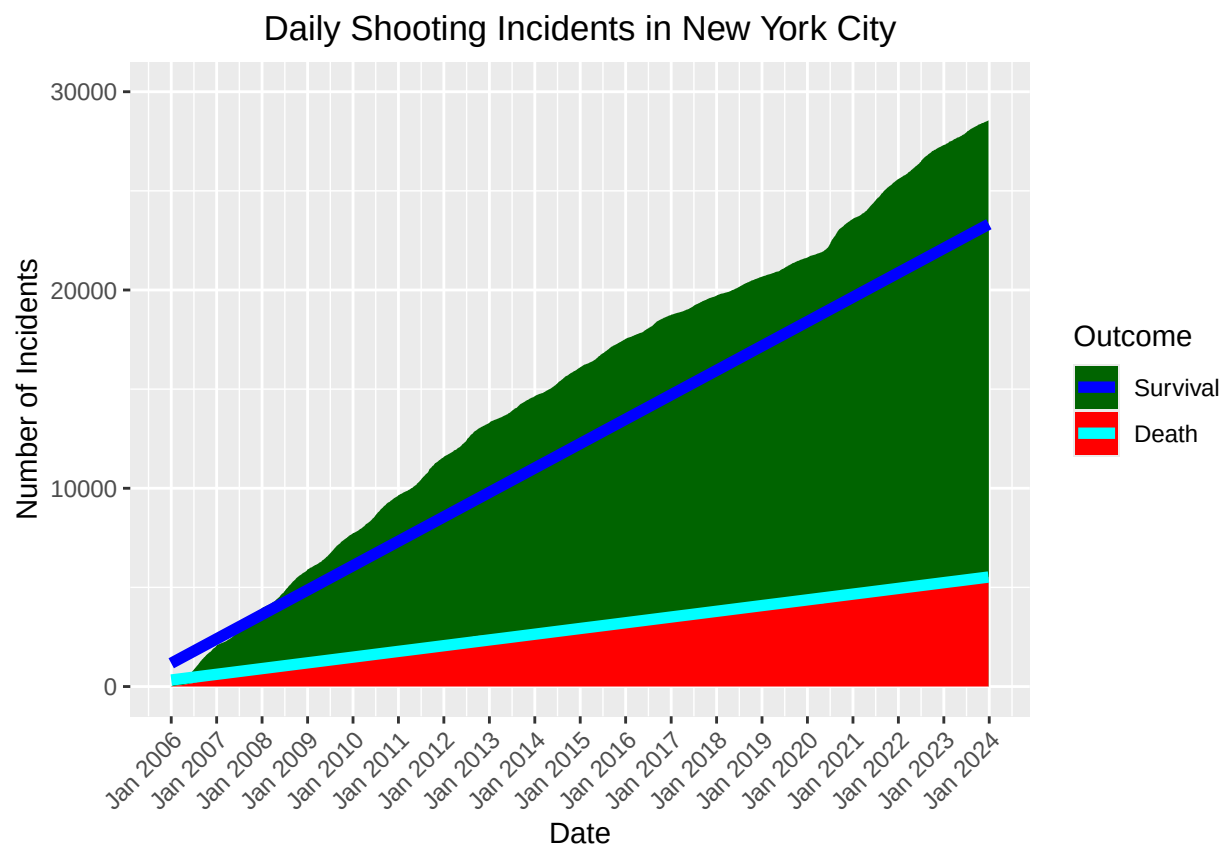


```
#Transformed cumulative data set nydeath_cuml that shows the total amount of incidents over time
nydeath_cuml <- nydeath %>%
  arrange(Date) %>%
  group_by(Death) %>%
  mutate(cumulative_count = cumsum(count))

#Condensed area chart of total amount of incidents over time
#Same stacked bar rule as the stacked bar chart
#Blue line: Linear line of best fit for survival incidents
#Cyan line: Linear line of best fit for death incidents
nydeath_cuml %>%
  ggplot(aes(x = Date, y = cumulative_count, fill = Death, color = Death)) +
  geom_area(position = "stack", color = NA) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 2) +
  labs(title = "Daily Shooting Incidents in New York City",
       x = "Date", y = "Number of Incidents",
       fill = "Outcome", color = "Outcome") +
  scale_x_date(date_labels = "%b %Y", date_breaks = "1 year") +
  scale_y_continuous(limits = c(0, 30000)) +
  scale_color_manual(values = c("TRUE" = "cyan", "FALSE" = "blue"),
                    name = "Outcome",
                    labels = c("TRUE" = "Death", "FALSE" = "Survival")) +
  scale_fill_manual(values = c("TRUE" = "red", "FALSE" = "darkgreen"),
                   name = "Outcome",
                   labels = c("TRUE" = "Death", "FALSE" = "Survival")) +
```

```
theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
theme(plot.title = element_text(hjust = 0.5))
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
#Transformed data set nyboro that shows the amount of incidents within each NYC borough
nyboro <- nyshooting %>%
  arrange(Boro)
```

```
nyboro <- nyboro %>%
  group_by(Boro, Death) %>%
  summarise(count = n(), .groups = "drop")
```

```
nyboro
```

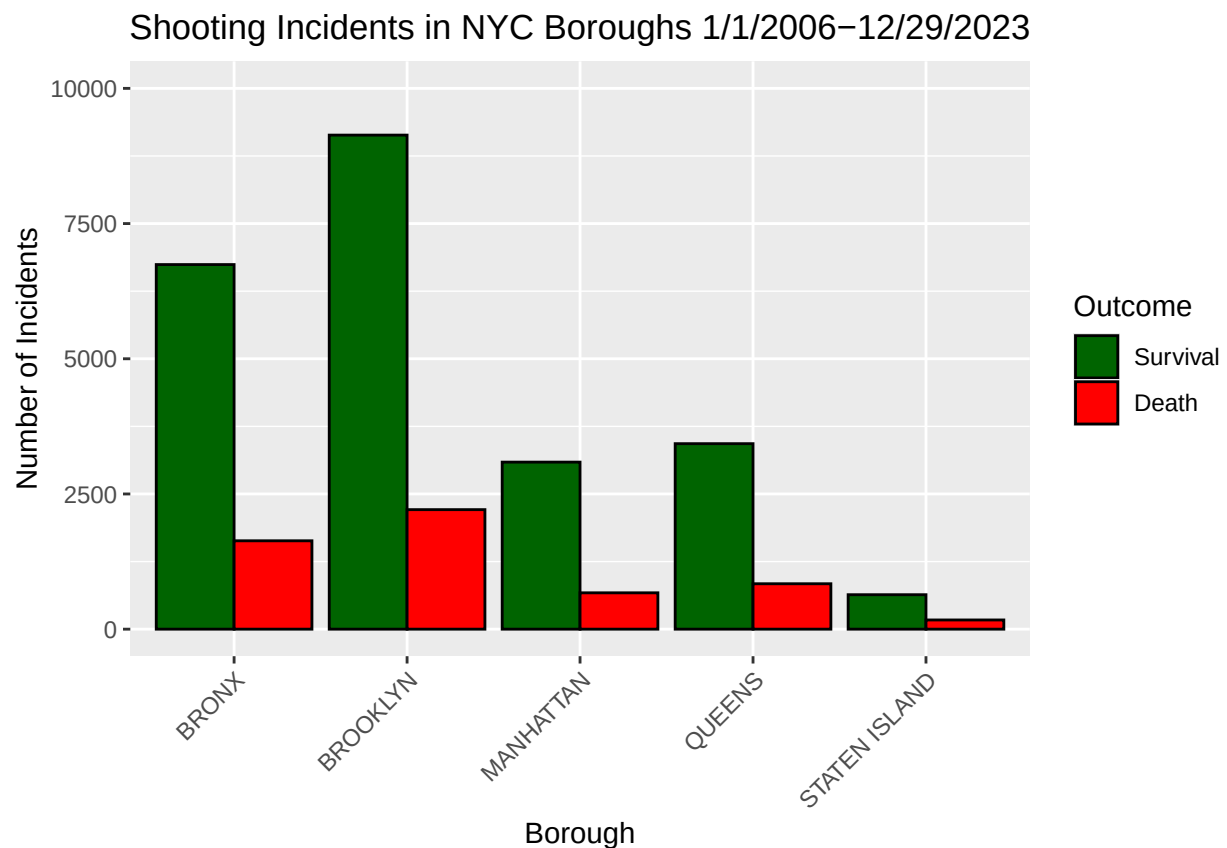
```
## # A tibble: 10 x 3
##   Boro      Death count
##   <fct>    <lg1> <int>
## 1 BRONX    FALSE  6742
## 2 BRONX    TRUE   1634
## 3 BROOKLYN FALSE  9136
## 4 BROOKLYN TRUE   2210
## 5 MANHATTAN FALSE  3089
## 6 MANHATTAN TRUE    672
```

```
## 7 QUEENS FALSE 3430
## 8 QUEENS TRUE 840
## 9 STATEN ISLAND FALSE 637
## 10 STATEN ISLAND TRUE 170
```

#Bar chart showing incidents in boroughs across the data's timeframe

```
nyboro %>%
```

```
  ggplot(aes(x = Boro, y = count, fill = Death)) +
  geom_col(position = "dodge", color = 'black') +
  scale_y_continuous(limits = c(0, 10000)) +
  scale_fill_manual(values = c("TRUE" = "red", "FALSE" = "darkgreen"),
                    name = "Outcome",
                    labels = c("FALSE" = "Survival", "TRUE" = "Death")) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(plot.title = element_text(hjust = 0.5)) +
  labs(title = "Shooting Incidents in NYC Boroughs 1/1/2006-12/29/2023",
       x = "Borough",
       y = "Number of Incidents",
       fill = "Death")
```



#Linear model of deaths as time passes

```
mod <- lm(count ~ Date, data = nydeath)
```

```
mod
```

```
##
```

```
## Call:
## lm(formula = count ~ Date, data = nydeath)
##
## Coefficients:
## (Intercept)      Date
##  5.0093509    -0.0001105
```

```
summary(mod)
```

```
##
## Call:
## lm(formula = count ~ Date, data = nydeath)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.557 -1.978 -0.918  0.989 33.029
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.0093509  0.2372181   21.117 < 2e-16 ***
## Date        -0.0001105  0.0000144   -7.672 1.88e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.637 on 8917 degrees of freedom
## Multiple R-squared:  0.006557,    Adjusted R-squared:  0.006445
## F-statistic: 58.85 on 1 and 8917 DF,  p-value: 1.877e-14
```

Data Findings: Brooklyn was the borough with the largest number of incidents, but that does not necessarily mean that it is the most dangerous borough to live in. There were a couple days where the number of incidents vastly outnumbered a calculated average, but for the most part the graph is consistent with incidents. The total number of incidents recorded from the clean data total to 23,034 incidents from January 1st, 2006 to December 29th, 2023.

Data Bias: There are differing law enforcement standards within each NYC borough, so the results may not show the true danger within each area. Death is a logical argument in the cleaned data set, so we do not know the extent of how much victims were injured before they died or survived. Race and age of attackers and victims were not variables of interest for what I wanted to uncover. Bias statements to consider when looking over the cleaned data and analysis.

```
write.csv(nyshooting, "nyshooting_cleaned.csv")
```

```
sessionInfo()
```

```
## R version 4.4.1 (2024-06-14 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
```

```

## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/Denver
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] lubridate_1.9.3 forcats_1.0.0  stringr_1.5.1  dplyr_1.1.4
## [5] purrr_1.0.2     readr_2.1.5    tidyr_1.3.1    tibble_3.2.1
## [9] ggplot2_3.5.1   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] Matrix_1.7-0      gtable_0.3.5      highr_0.11        compiler_4.4.1
## [5] tidyselect_1.2.1  splines_4.4.1     scales_1.3.0      yaml_2.3.10
## [9] fastmap_1.2.0     lattice_0.22-6    R6_2.5.1          labeling_0.4.3
## [13] generics_0.1.3    knitr_1.48        munsell_0.5.1     pillar_1.9.0
## [17] tzdb_0.4.0        rlang_1.1.4       utf8_1.2.4        stringi_1.8.4
## [21] xfun_0.51         timechange_0.3.0  cli_3.6.3         mgcv_1.9-1
## [25] withr_3.0.1       magrittr_2.0.3    digest_0.6.36     grid_4.4.1
## [29] rstudioapi_0.16.0 hms_1.1.3         nlme_3.1-164      lifecycle_1.0.4
## [33] vctrs_0.6.5       evaluate_0.24.0   glue_1.7.0        farver_2.1.2
## [37] fansi_1.0.6       colorspace_2.1-1  rmarkdown_2.28    tools_4.4.1
## [41] pkgconfig_2.0.3   htmltools_0.5.8.1

```